Manual

Rule-Based Machine Assignment

HLS-RBM 8.2

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Contents

1 Rule-Based Machine Assignment

Purpose

Priority rules are used in order processing for detailed planning (and sequencing) once an order has been released. With the aid of priority rules, detailed planning specifies how released orders are assigned to workstations. This becomes particularly important when several orders are lined up at a workstation awaiting processing, and thus competing for resources.

Based on a queuing model, a random number of resources (generally machines / workplaces) is combined, from which a queue of operations ready for planning is generated. Each queue is given a priority rule to identify the highest priority operation (so-called contender) in the queue.

Implementation notes

You use the function package if the queue sequence should be built based on one of the priority rules listed below:

Integration

The priority rules are used as part of automatic planning / assignment in order to specify the sequence with regard to the next highest-priority orders / operations.

Features

- Planning strategies controlling automatic machine assignment based on standard strategies:
    - Longest processing time remaining (GRB)
    - Shortest operation time (KOZ)
    - Shortest processing time remaining (KRB)
    - Longest operation time (LOZ)
    - Lowest setup times (UK)
- Selection of sorting rules when defining planning options

2 Automatic Assignment

Purpose

You can use the function *Automatic assignment* to automatically perform a planning scenario, which is defined via planning profile and planning horizon.

The function is used to plan orders and the included operations to capacities. The system uses assessment criteria – the so-called *Assignment strategy* – to sort the operations and to distribute the operations to the workplaces of the group.

Requirements

Before you start the detailed planning, you must define and create the relevant Planning profiles that integrate aspects of organization, planning procedure and competencies.

2.1 Automatic assignment

The automatic assignment, also called capacity planning, is used to plan orders or their operations to capacities. To meet the demand of each operation, the system selects a resource or a combination of resources from the resources available. As a result of planning, the operation is assigned a start and end time (planned start and planned end).

To calculate the planned duration of the operation, the system uses the workplace/machine that meets the main demand plus the dynamic setup time between the preceding operation and the operation itself.

The workplace/machine is the resource that meets the "main demand". The alternatives to meet the demand are the workplaces/machines of the same group (the group the operation has been assigned to). The assignment of a workplace to a group is configured in the master data.

Production resources and tools are used to meet the further demand. They are stored for the operation.

If Production variants are used, the combination of workplace and resource meets the relevant demand. Here, a resource is specified in the Tool and Resource Management (WRM) and assigned to a production variant.

An automatic planning run is performed in the following steps:

1. The order of assignment is specified via sorting of the operations that must be planned ("queue") according to the defined ***planning strategy*** (*priority rule*).

   The orders that must be planned are sorted according to their priority and placed in a queue. To identify the priorities, different rules are available.

2. The operations are distributed among the available workplaces according to the specified procedure (***control of capacity selection***).

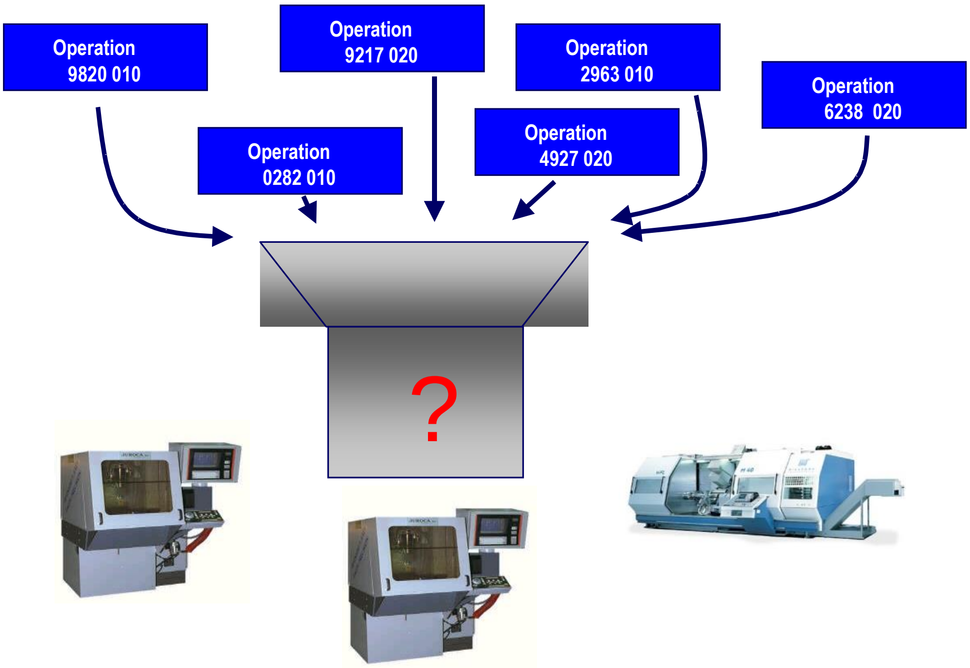
The following rule applies to **specify the planned date** of an operation that has been assigned to a machine via capacity selection:
- If the planning variant includes the active option *Displacem. to the left* (left shift) for the planning horizon, then the earliest possible date within the planning horizon is selected as planned start. With this planned start, the capacity is available and the relationships between operations are respected and free of conflicts. - If the option *Displacem. to the left* (left shift) is not active in the planning variant, the system tries to schedule the operation at its **scheduled start time**. In case of a forward scheduling, the start time is the **earliest start**, in case of a backward scheduling, it is the **latest start** of the operation. If no capacity is available at that time, the next later point in time is identified when the required capacity is available. With backward scheduling, the operation is then scheduled "too late". Optionally, the option IGNORE_TIMING_DIRECTION can be set to "J" in the INI configuration. With this configuration and in case of backward scheduling, the automatic planning searches for a start date beginning with the **earliest start**. To identify the start date, the planned machine must provide the required capycity.

The result of the automatic assignment is a planning scenario that can actually be realized. This means that the planning is free of capacity overloads or conflicts because of relationships. If these conditions must be fulfilled, it might not be possible to schedule all operations in due time. It is therefore possible that some operations must be planned with a delay. This depends on the planning situation, the available resources and the selected planning strategy. If the capacity required for an operation is not available until the end of the planning horizon and with respect of the situation and the respective configuration, then this operation remains in the pool of groups also after an automatic assignment.

2.1.1 Specification of the assignment order

The automatic assignment is based on a queue concept. The workplaces assigned to a capacity group are regarded as a unit, in front of which a queue is formed, i.e. the operations to be planned line up in front of this unit. Each queue is given a priority rule to identify the highest priority operation (so-called contender) in the queue.

The user can specifically set queue sorting (assignment strategy) when the system is customized and therefore adapt it to the respective requirements. Each operation is weighted with the corresponding factors based on the assignment strategy. The operations are then sorted accordingly and then distributed to the workplaces.

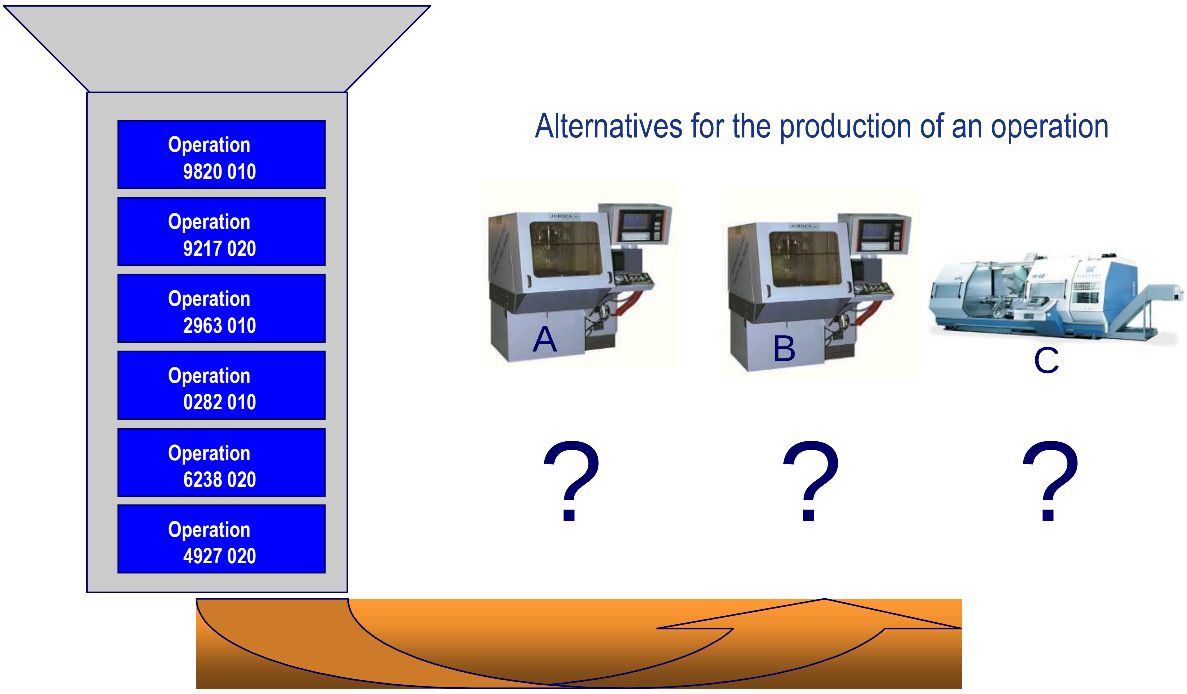


Various bottleneck and scheduling resources can be defined for scheduling within the planning horizon (the specification is made in the group). When you schedule operations within the planning horizon, resources are checked, i.e. the system checks availability of all resources not only of workplaces.

Apart from the required demand and the possible elements that meet the demand, capacity planning also takes into account further restrictions, such as maximum time intervals between two operations to be synchronized.

2.1.2 Controlling capacity selection

How operations are distributed to the individual workplaces can be controlled using the adjustable capacity selection in the planning variants.



The following alternatives are available:

The first free capacity

The capacity that is available first in terms of time.

Capacity having the most "capacity"

Selects the capacity that provides the most available capacity with respect to the planning horizon.

Capacity having the least "capacity"

The capacity is selected which, with respect to the planning horizon, provides the least capacity.

Shortest processing time

When attempting to dispatch operations, the order in which capacities are selected is based on the shortest processing time (while taking into account performance level).

Please note: This is of particular significance in conjunction with alternative production methods (production variants), since these provide certain values used in calculating process times.

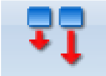
Shortest setup time

When attempting to dispatch operations, the order in which capacities are selected is based on the shortest setup time. Dynamic setup times based on the increases or reductions in setup time defined in the setup change matrix are taken into consideration.

Constraints within the order network are appropriately taken into consideration during automatic scheduling. Operations that are already running are not rescheduled. If an assignment is not possible because there are overlaps with the predecessor or successor operations, for example, the operation is not scheduled and appears in a list at the end of automatic planning.

During automatic assignment, fixed operations that are still set in the past are moved to the right and set to "now" plus planning lead time at the earliest. Fixed operations in the future are not modified, but remain scheduled as they are.

2.1.3 Complete assignment of planning board

Automatic assignment of the entire planning board is accessed using the icon  in the toolbar. A dialog opens from which the following options can be selected:

Schedule unplanned operations only (pre-assignment)

Using this option operations that are in the pool of groups are weighted according to the current assignment strategy and distributed to/ scheduled for the workplaces of the group. Operations that have already been scheduled remain scheduled in their position (workplace, date).

If an assignment is not possible because there are overlaps with the predecessor or successor operations, for example, the operation is not scheduled and appears in a list at the end of automatic planning.

Reschedule entire planning scenario

If all operations are to be rescheduled, select option "Reschedule all". In this case, all workplaces for operations already scheduled, and that are not fixed or running, are scheduled first, that is moved to the pool of groups before automatic assignment takes place.

After this has been confirmed, the separate workplaces are now assigned. In the process, all operations in the pool of groups are weighted according to the current assignment strategy. These operations are then distributed/ dispatched to the workplaces in the group.

If an assignment is not possible because there are overlaps with the predecessor or successor operations, for example, the operation is not dispatched and appears in a list at the end of automatic planning.

2.2 Planning strategies/priority rules

A planning strategy/ priority rule specifies a priority code (PKZ) for each of the operations transferred to the queue.

In addition to the predefined strategies in the system - rule-based machine assignment - user-defined strategies are also possible (target-oriented rules and sorting). Whether or not individual planning algorithms can be used for this purpose depends on the license. Planning algorithms are selected through selecting a planning variant.

2.2.1 Rule-based machine scheduling

Lowest setup costs

The setup costs equal the sum of setup time + teardown time.

The operation in the queue having the lowest setup costs for the machine in question gets the highest priority (dynamic setup times are not included; these times are used with the capacity selection).

Shortest operation time

Operation time = setup time + processing time + machine teardown time

The operation in the queue having the shortest operation time gets the highest priority.

Shortest remaining processing time

The operation in the queue having the shortest remaining processing time gets the highest priority. The remaining processing time is calculated using the remaining run time formula stored for the operation.

Longest operation time

Operation time = setup time + processing time + machine teardown time

The operation in the queue having the longest operation time gets the highest priority.

Longest remaining processing time

The operation in the queue having the longest remaining processing time gets the highest priority. The remaining processing time is calculated using the remaining run time formula stored for the operation.

2.2.2 Variable machine assignment

Planning strategy for controlling automatic machine assignment based on customer-specific sorting criteria (sorting rules).

The sorting rule calculates a priority number for each of the operations presented. It works more or less like a sifter in which ultimately only the highest priority operation in the queue is trapped. If, for example, only one operation is left over after the first sorting criterion is processed, this operation is scheduled as the next one and the other sorting criteria are not further considered.

The configuration used to create sorting rules is described here. The possible sorting criteria are also described there.

By default, automatic assignment is carried out in the following sequence:

	Sequence	Sorting	
1	Scheduled start (date) of operation	Ascending	
2	Priority of operation	Descending	
3	Order buffer time (from scheduling)	Ascending	
4	Order index of the order	Descending	
5	HYDRA order number (order/OP number)	Ascending	

2.2.3 Target-oriented machine assignment

Planning strategy for controlling automatic machine assignment based on weighted targets.

This planning strategy involves indexing the operations to be scheduled according to a variable weighting and then sorting the queue according to this index. In contrast to priority rules, the actual target is weighted in this case, and assignment is then aligned with the target.

The configuration used to create sorting rules is described here. The possible sorting criteria are also described there.

2.3 Other notes

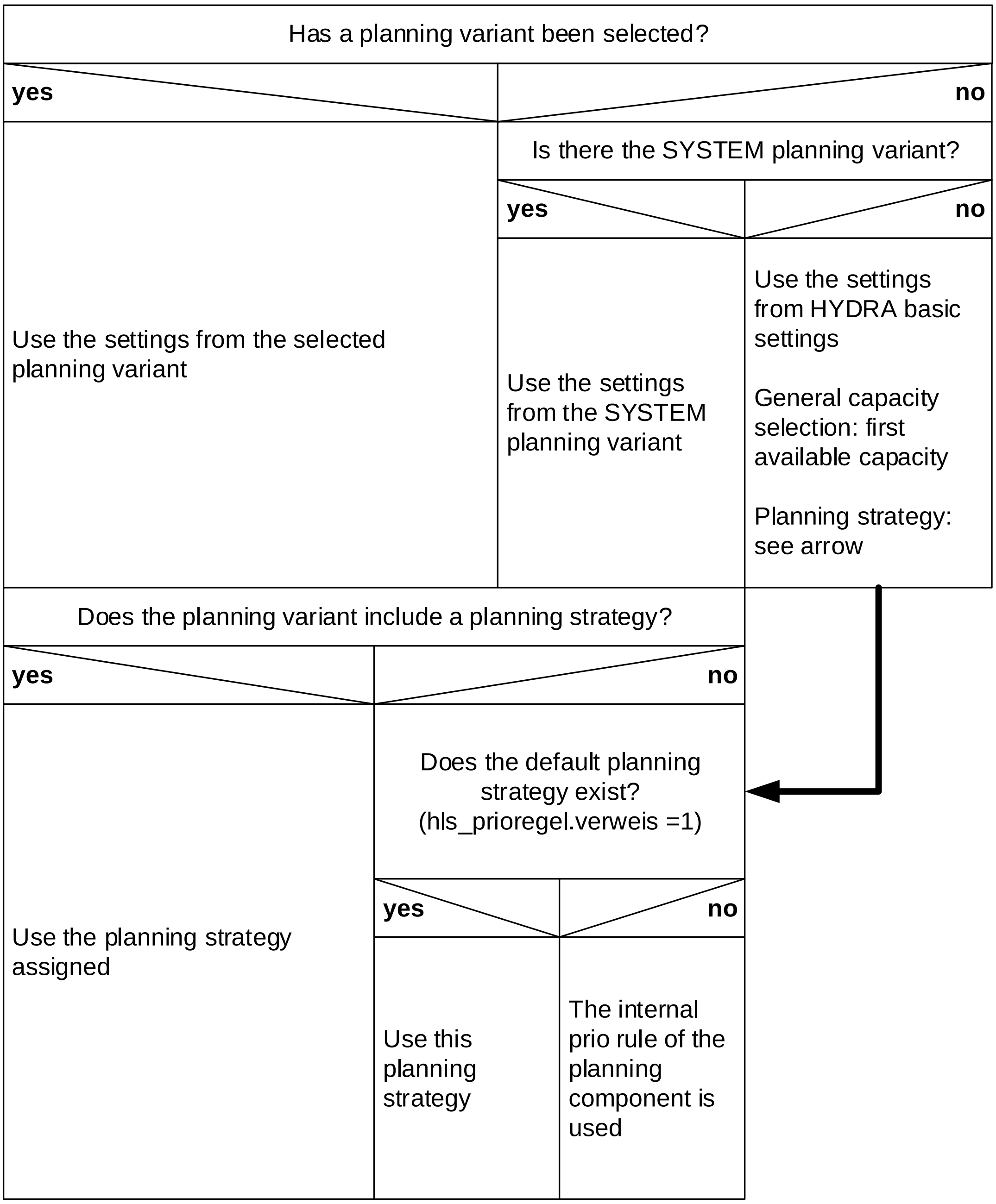
Please note the following:

Taking wait time into consideration

The wait time defined in the operation is NOT taken into consideration in planning (automatic assignment or manual assignment). This wait time is considered being a wait time for planning purposes that is thus only considered as buffer time when it comes to scheduling the lead time.

Planning variants and basic settings

You can make different settings for the time horizons and the automatic assignment in the *Basic settings* and in the *Planning variants*. If you do not select a planning variant in the *Graphic planning*, the basic settings are used. The following graphic shows the logic applied for the use of the different settings.



The internal priority rule (sorting rule) of the planning component is as follows:

	Sequence	Sorting	
1	Scheduled start (date) of operation	Ascending	
2	Priority of operation	Descending	
3	Order buffer time (from scheduling)	Ascending	
4	Order index of the order	Descending	
5	HYDRA order number (order/OP number)	Ascending	